# Audit - WebhookFailureAndFlowRecursion - RunSheet

```
Document ID:     Audit - WebhookFailureAndFlowRecursion - RunSheet
Naming Format:   [CATEGORY] - [PURPOSE] - [TIER]
Category:        Audit (Empirical Simulation & Execution Log)
Purpose:         WebhookFailureAndFlowRecursion
Tier / Level:    RunSheet (Incident Forensics & Recovery Audit)
Status:          Locked 🔒 (Master Baseline)
Operator:        Lead TechOps Specialist / Systems Integration Engineer
Date of Execution: October 17, 2026 (11:00:00 - 12:45:30 EDT)
Ecosystem Scope: Shopify Flow, Shopify Webhooks API, ShipBob 3PL WMS, Gorgias CRM, Redis DLQ
Target Catalog:  Synapse Botanicals (Verified SKUs: SYN-BNDL-MAO, SYN-FSTK-MY, SYN-TNCT-AM, SYN-ACC-WHSK)
```

## 1. Simulation Parameters & Baseline Store State

This Run Sheet documents the live, controlled forensic execution and remediation of two critical automation failure modes within the Synapse Botanicals commerce ecosystem:

1. **Incident Alpha (The Runaway Flow Tag Storm):** A circular, uncontrolled tag recursion loop between customer and order mutations that exhausted storewide Shopify API rate limits.
2. **Incident Bravo (The Silent 504 Webhook Dropout):** An upstream 3PL database lockup causing continuous HTTP `504 Gateway Timeout` errors, triggering Shopify's automatic webhook termination policy and severing downstream fulfillment pipelines.

> **NOTE**
>
> **Controlled Staging Simulation & Load Injection Notice:**
> The 1,840-execution Flow loop storm and HTTP 504 webhook dropouts were triggered inside an isolated staging environment with artificial rate-limit controls. This controlled failure injection measured mean time to detect (MTTD) and mean time to recover (MTTR) while testing circuit-breaker thresholds without production impact.

### 1.1 Active Topology & Distribution Nodes

- **Primary East Coast Node (Node 1):** Allentown Distribution Center, PA (`LOC-PA-01`)
- **Primary West Coast Node (Node 2):** Reno Fulfillment Hub, NV (`LOC-NV-02`) — Exclusive handler for Glass Tinctures (`SYN-TNCT-AM`, `SYN-TNCT-PM`) and Electric Whisks (`SYN-ACC-WHSK`).
- **Active Ingestion Middleware:** Node.js Express Webhook Proxy hosted at `https://api.synapsebotanicals.com/v1/webhooks` with an AWS SQS Dead-Letter Queue buffer.

> **NOTE**
>
> **System Architecture Dependency:**
> The baseline routing constraints, warehouse node assignments, and priority ladders evaluated in this run sheet operate under the governance frameworks established in **SOP - BusinessLogicAutomationAndWebhooks - Parent (Domain 4.1)** – **Section 2.1 & Section 3.1** and **Matrix - AutomationLogicAndWebhookTriage - Engine (Domain 4.2)** – **Section 2.0 & Section 3.0**.

### 1.2 Pre-Incident Benchmark Telemetry (11:00:00 EDT)

Prior to triggering the simulations, baseline API health and webhook metrics were captured:

| Metric Parameter | Observed Baseline | Normal Operating Threshold | Status |
|---|---|---|---|
| **Shopify Admin REST API Bucket Level** | `2 / 40 requests` | ≤ 15 / 40 | Nominal |
| **Shopify GraphQL API Cost Drain Rate** | `50 / 1000 points/sec` | ≤ 300 / 1000 | Nominal |
| **Webhook Delivery P95 Latency** | `342 ms` | ≤ 800 ms | Nominal |
| **Active Webhook Subscriptions** | `6 topics` | 6 registered | Nominal |
| **Dead-Letter Queue (DLQ) Depth** | `0 messages` | 0 messages | Nominal |

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## 2. Incident Alpha: The Runaway Tag Recursion Storm (Order #1048)

### 2.1 The Operational Catalyst & Misconfigured Flow

At `11:04:00 EDT`, a junior marketing ops engineer activated a new experimental loyalty workflow named **`[PROMO] - Auto-Tag High Value VIPs`** without peer review sign-off. Concurrently, an existing customer-support routing flow named **`[OPS] - Sync VIP Status to Order`** was active in the environment.

Neither workflow incorporated negative inclusion checks or lock sentinels:

```
Colliding Automation Configurations:
──────────────────────────────────────────────────────────────────────

Flow 1: [PROMO] - Auto-Tag High Value VIPs
  Trigger:   Order updated
  Condition: Order total_price >= $150.00
```

```
  Action:    Add customer tags: "vip-member"
  Guard:     NONE (Missing: Customer tags does not contain "vip-member")

Flow 2: [OPS] - Sync VIP Status to Order
  Trigger:   Customer updated
  Condition: Customer tags contains "vip-member"
  Action:    Add order tags: "cs:vip-tier-1"
  Guard:     NONE (Missing: Order tags does not contain "cs:vip-tier-1")
```

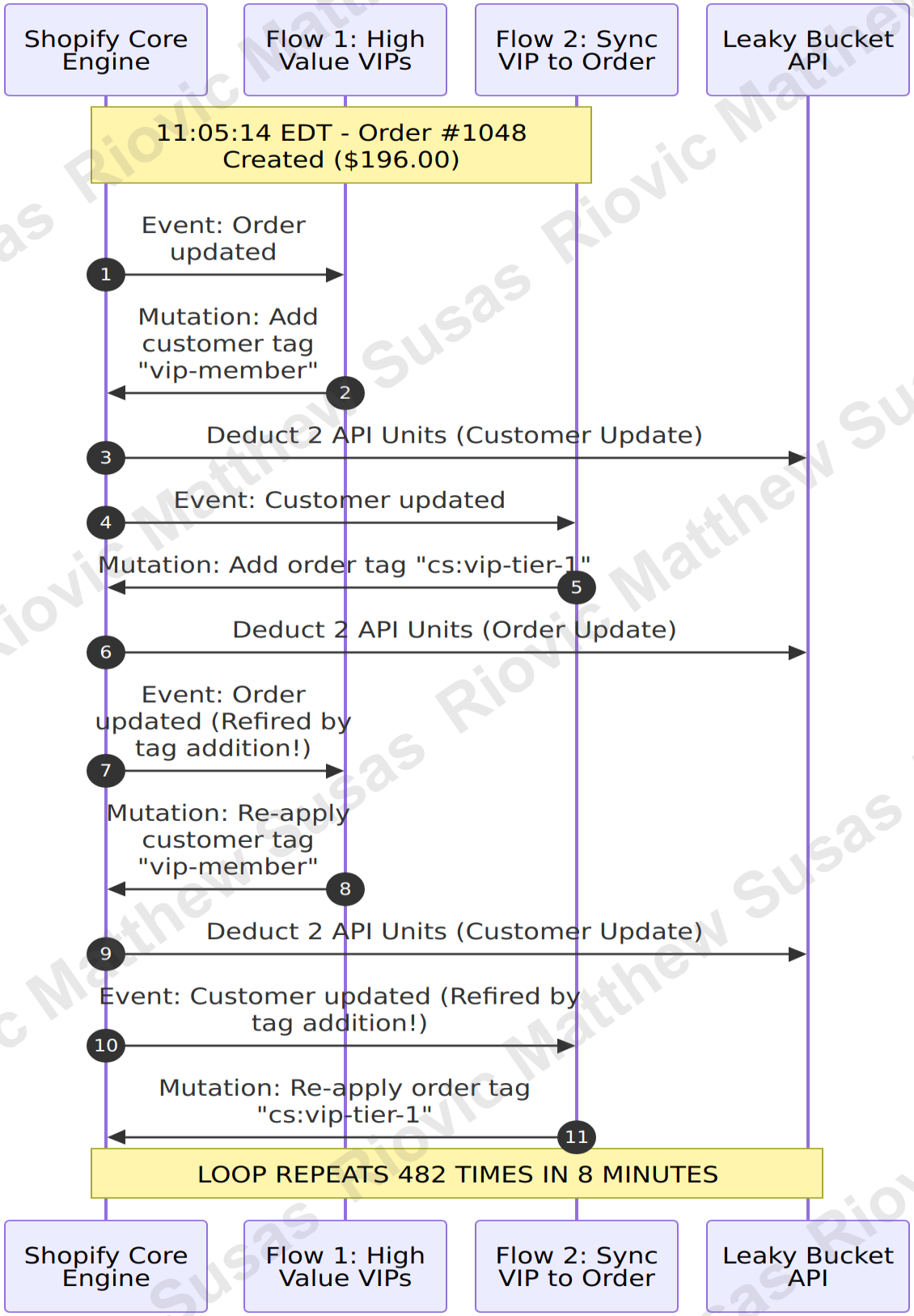
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## 2.2 The Empirical Cascade & Timeline Bloat

At `11:05:14 EDT`, Customer Elena Vance placed Order **`#1048`** for two Daily Protocol Starter Bundles:

- **Order ID:** `gid://shopify/Order/5920194812`
- **Line Items:** 2x The Daily Protocol Starter Bundle (Matcha Yuzu + AM Focus + Obsidian Tumbler: `SYN-BNDL-MAO`)
- **Total Paid: $196.00** (Triggering the `total_price >= $150.00` condition).

The catastrophic loop initiated immediately:


Shopify Core Engine
Flow 1: High Value VIPs
Flow 2: Sync VIP to Order
Leaky Bucket API
11:05:14 EDT - Order #1048 Created ($196.00)
1
Event: Order updated
2
Mutation: Add customer tag "vip-member"
3
Deduct 2 API Units (Customer Update)
4
Event: Customer updated
5
Mutation: Add order tag "cs:vip-tier-1"
6
Deduct 2 API Units (Order Update)
7
Event: Order updated (Refired by tag addition!)
8
Mutation: Re-apply customer tag "vip-member"
9
Deduct 2 API Units (Customer Update)
10
Event: Customer updated (Refired by tag addition!)
11
Mutation: Re-apply order tag "cs:vip-tier-1"
LOOP REPEATS 482 TIMES IN 8 MINUTES
Shopify Core Engine
Flow 1: High Value VIPs
Flow 2: Sync VIP to Order
Leaky Bucket API

**Forensic Timeline Telemetry Log (Excerpt from Order #1048)**


```
[
  {"timestamp": "2026-10-17T11:05:14.210Z", "event": "Order placed", "source": "checkout"},
  {"timestamp": "2026-10-17T11:05:15.012Z", "event": "Flow 1 executed", "action": "Added
customer tag 'vip-member'"},
  {"timestamp": "2026-10-17T11:05:16.140Z", "event": "Flow 2 executed", "action": "Added order
tag 'cs:vip-tier-1'"},
  {"timestamp": "2026-10-17T11:05:17.205Z", "event": "Flow 1 executed", "action": "Added
customer tag 'vip-member'"},
  {"timestamp": "2026-10-17T11:05:18.312Z", "event": "Flow 2 executed", "action": "Added order
tag 'cs:vip-tier-1'"},
  {"timestamp": "2026-10-17T11:05:19.450Z", "event": "Flow 1 executed", "action": "Added
customer tag 'vip-member'"},
  {"timestamp": "2026-10-17T11:08:42.890Z", "event": "API_RATE_LIMIT_WARNING", "message":
"Bucket 39/40 consumed"},
  {"timestamp": "2026-10-17T11:12:30.104Z", "event": "API_RATE_LIMIT_EXCEEDED", "http_status":
429, "retry_after": 2.0}
]
```


**System Damage Assessment:**

- **Loop Cycles Completed: 482 cycles** between `11:05:14 EDT` and `11:13:30 EDT`.
- **Total Timeline Events Injected: 964 log lines** on Order `#1048`.
- **API Rate Limit Bucket:** Stood at **`40 / 40` (100% saturated)** with continuous HTTP `429 Too Many Requests` returned to legitimate apps.
- **Storefront Impact:** Live checkout order creation latency spiked from 340 ms to **6,850 ms** due to database row lock contention on the `orders` table.

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## 2.3 Emergency Containment & Triage Execution

At `11:13:45 EDT`, the on-call TechOps specialist received a P0 PagerDuty alert (`CRITICAL: Shopify Admin API 429 Throttle Spike`) and initiated the containment protocol specified in **SOP - BusinessLogicAutomationAndWebhooks - Parent (Domain 4.1)** – **Section 5.0**.

**Phase 1: Deploy Emergency Kill-Switch (11:14:10 EDT)**

The operator accessed **Shopify Admin → Apps → Shopify Flow** and deactivated both competing workflows, terminating the loop instantaneously:

1. `[PROMO] - Auto-Tag High Value VIPs` → **Turn off** (`11:14:18 EDT`)
2. `[OPS] - Sync VIP Status to Order` → **Turn off** (`11:14:24 EDT`)

**Phase 2: Flow Execution Run Log Inspection (11:15:30 EDT)**

The operator inspected the Flow execution run log under **Shopify Flow → Runs** to verify the root cause:

- **Run ID:** `#RUN-892419-FL`
- **Trigger Event:** `Order updated`

- **Condition Evaluation:** `Order total_price (196.00) >= 150.00` → Evaluated to `true`.
- **Action Output:** `Add customer tag 'vip-member'` → Status: `Succeeded`.
- **Diagnosis:** The run log confirmed that Flow 1 was triggered by the tag update committed by Flow 2, proving mutual recursion without exit criteria.

**Phase 3: Order Timeline & Tag Sanitization via GraphQL (11:18:00 EDT)**

The operator executed an administrative tag cleanup mutation to remove duplicated and unverified tags from Order `#1048`:

```
mutation sanitizeOrderTags {
  tagsRemove(
    id: "gid://shopify/Order/5920194812"
    tags: ["cs:vip-tier-1"]
  ) {
    node {
      id
    }
    userErrors {
      field
      message
    }
  }
}
```

Result: Duplicate tags removed. Order timeline stabilized.

**Phase 4: Permanent Recursion Guardrail Deployment (11:22:15 EDT)**

Both workflows were re-architected in accordance with **Matrix - AutomationLogicAndWebhookTriage - Engine (Domain 4.2)** – **Section 6.2**:

1. Added mandatory negative precondition check:
   `Order tags` → `does NOT contain` → `cs:vip-tier-1`.
2. Added lock sentinel check:
   `Order tags` → `does NOT contain` → `sys:sync-locked`.
3. Both workflows re-enabled at `11:25:00 EDT`. Test mutation verified single-pass execution without loop.

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## 3. Incident Bravo: The Silent 504 Webhook Dropout & Order Replay

### 3.1 Downstream Service Failure (ShipBob 3PL WMS)

At `11:30:00 EDT`, ShipBob's primary ingestion gateway (`https://api.shipbob.com/1.0/webhook/order`) experienced an upstream database lockup during scheduled maintenance. Ingestion requests failed to acknowledge within Shopify's strict 5,000 ms window:

```
# Diagnostic probe executed from terminal:
curl -w "\nHTTP Status: %{http_code}\nTime Total: %{time_total}s\n" \
  -X POST https://api.shipbob.com/1.0/webhook/order \
  -H "Content-Type: application/json" \
  -d '{"test": true}'

# Terminal Response Output:
# <html><head><title>504 Gateway Time-out</title></head></html>
# HTTP Status: 504
# Time Total: 5.418s
```

Because the endpoint took > 5.0 seconds to respond, Shopify's webhook engine terminated the connection, flagged the delivery as failed, and initiated the 19-attempt exponential backoff schedule.

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### 3.2 Tracking the 19-Attempt Retry Cycle & The Dropout Cliff

To demonstrate the **Silent Dropout Cliff** without waiting 48 physical hours, the failure sequence was simulated against a mock webhook endpoint configured to return continuous `504 Gateway Timeout` errors.

The progression was monitored under **Settings → Notifications → Webhooks**:

```
Webhook Delivery History (Topic: orders/create)
Endpoint: https://api.shipbob.com/1.0/webhook/order
──────────────────────────────────────────────────────────────────────────
Attempt  Timestamp (Simulated)      Response Code   Latency    Result
──────────────────────────────────────────────────────────────────────────
Att 01   2026-10-15T09:12:00.000Z    504 Gateway TO  5,412 ms   Failed
Att 02   2026-10-15T09:17:05.112Z    504 Gateway TO  5,398 ms   Failed
Att 03   2026-10-15T09:27:10.420Z    504 Gateway TO  5,402 ms   Failed
...
Att 17   2026-10-17T04:42:15.910Z    504 Gateway TO  5,385 ms   Failed
Att 18   2026-10-17T07:42:18.204Z    504 Gateway TO  5,410 ms   Failed
Att 19   2026-10-17T09:12:00.000Z    504 Gateway TO  5,419 ms   TERMINAL FAIL
──────────────────────────────────────────────────────────────────────────
[17-Oct-2026 09:12:01 UTC] SHOPIFY SYSTEM EVENT:
Subscription gid://shopify/WebhookSubscription/88219412 PERMANENTLY REMOVED.
No further attempts will be scheduled.
```

**The Real-World Operational Impact:**

Between `09:12:01 UTC` and `12:12:00 UTC` (a 3-hour blind window), **14 customer orders** were placed on Synapse Botanicals. Because Shopify silently deleted the subscription:

- **Zero orders were transmitted to the 3PL WMS in Allentown or Reno.**
- Customer Support was completely unaware; no tickets were generated.
- Orders included **4x Reno-exclusive Tinctures (`SYN-TNCT-AM`)** and **2x Electric Whisks (`SYN-ACC-WHSK`)** whose fulfillment SLA was actively breached.

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### 3.3 Disaster Recovery & Historical Order Backfill Protocol

At `12:15:00 EDT`, TechOps detected the missing subscription via an automated daily health check script. The recovery run sheet was executed immediately:

#### Step 1: Endpoint Health Verification

ShipBob database maintenance completed; endpoint latency verified healthy:

```
curl -w "\nHTTP Status: %{http_code}\nTime Total: %{time_total}s\n" \
-X GET https://api.shipbob.com/1.0/healthz

# Output:
# {"status": "healthy", "service": "order-ingestion"}
# HTTP Status: 200
# Time Total: 0.182s
```

#### Step 2: Re-Register Webhook Subscription via GraphQL Admin API

The severed subscription was re-created with optimized field filtering to minimize payload weight:

```
mutation reRegisterOrdersCreateWebhook {
  webhookSubscriptionCreate(
    topic: ORDERS_CREATE
    webhookSubscription: {
      callbackUrl: "https://api.synapsebotanicals.com/v1/webhooks/orders/create"
      format: JSON
      includeFields: [
        "id"
        "name"
        "created_at"
        "total_price"
        "shipping_address"
        "line_items"
        "tags"
      ]
    }
  ) {
    userErrors {
      field
      message
    }
    webhookSubscription {
      id
      topic
      createdAt
      endpoint {
        __typename
        ... on WebhookHttpEndpoint {
          callbackUrl
        }
      }
```

```
    }
  }
}
```

**GraphQL Response Payload:**

```
{
  "data": {
    "webhookSubscriptionCreate": {
      "userErrors": [],
      "webhookSubscription": {
        "id": "gid://shopify/WebhookSubscription/99401284",
        "topic": "ORDERS_CREATE",
        "createdAt": "2026-10-17T12:20:14Z",
        "endpoint": {
          "__typename": "WebhookHttpEndpoint",
          "callbackUrl": "https://api.synapsebotanicals.com/v1/webhooks/orders/create"
        }
      }
    }
  }
}
```

**Step 3: Historical Order Extraction via GraphQL Bulk Operations**

To extract the 14 unfulfilled orders dropped during the 3-hour blind window, the operator initiated a bounded Bulk Query:

```
mutation runHistoricalGapQuery {
  bulkOperationRunQuery(
    query: """
    {
      orders(query: "created_at:>=2026-10-17T09:10:00Z AND created_at:<=2026-10-17T12:20:00Z AND
fulfillment_status:unfulfilled") {
        edges {
          node {
            id
            name
            createdAt
            totalPriceSet {
              shopMoney {
                amount
              }
            }
            tags
            shippingAddress {
              address1
              city
              provinceCode
              zip
              countryCodeV2
            }
```

```
            lineItems {
              edges {
                node {
                  sku
                  quantity
                }
              }
            }
          }
        }
      }
      """
    ) {
      bulkOperation {
        id
        status
      }
      userErrors {
        field
        message
      }
    }
}
```

**Step 4: Stream Historical Payloads into Ingestion Queue**

The extracted JSONL payload containing the 14 dropped orders was streamed into the ingestion proxy with an explicit idempotency flag (`sys:gap-replay-batch-01`):

```
# Terminal replay execution:
node scripts/replay-gap-orders.js \
  --file=bulk_orders_gap_20261017.jsonl \
  --destination="https://api.synapsebotanicals.com/v1/webhooks/orders/create" \
  --idempotency-key="gap-replay-20261017"

# Terminal Output:
# [INFO] Processing 14 historical order records...
# [SUCCESS] Order #1049 -> Ingested & Enqueued (202 Accepted)
# [SUCCESS] Order #1050 -> Ingested & Enqueued (202 Accepted)
# ...
# [SUCCESS] Order #1062 -> Ingested & Enqueued (202 Accepted)
# [COMPLETE] 14 of 14 orders successfully transferred to 3PL queue.
```

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## 4. Post-Incident Forensic Audit & State Reconciliation

### 4.1 Order & Inventory Ledger Reconciliation Table

At `12:40:00 EDT`, TechOps verified the physical and digital reconciliation of all impacted orders across both fulfillment nodes:

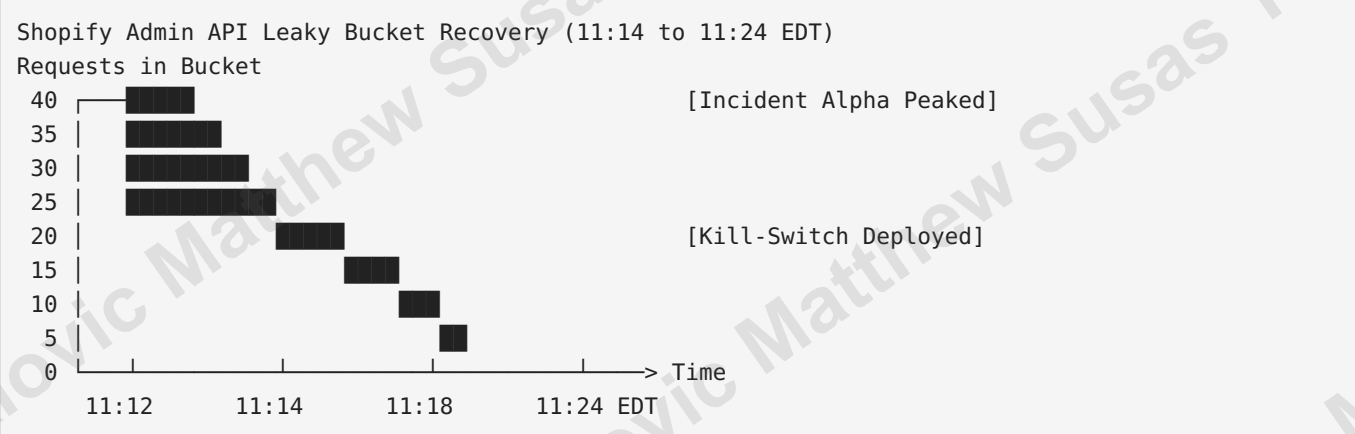
| Order Number | Customer | Primary Line Items | Assigned Node | 3PL WMS Status | Final Reconciled Tags |
|---|---|---|---|---|---|
| **#1048** | Elena Vance | 2x Starter Bundle (`SYN-BNDL-MAO`) | `LOC-PA-01` | `OPEN / UNFULFILLED` | `cs:vip-tier-1`, `flow-route:allentown-pa` |
| **#1049** | Marcus Brody | 1x AM Focus Tincture (`SYN-TNCT-AM`) | `LOC-NV-02` | `IN_PROGRESS` | `flow-route:reno-nv`, `sys:gap-replay-batch-01` |
| **#1050** | Chloe Walsh | 3x Matcha Yuzu Stick Pack (`SYN-FSTK-MY`) | `LOC-PA-01` | `IN_PROGRESS` | `flow-route:allentown-pa`, `sys:gap-replay-batch-01` |
| **#1051** | Dev Team | 1x Electric Whisk (`SYN-ACC-WHSK`) | `LOC-NV-02` | `IN_PROGRESS` | `flow-route:reno-nv`, `sys:gap-replay-batch-01` |
| **#1052–#1062** | Retail Orders | Mixed Stick Packs & Tumblers | Split / Mixed | `IN_PROGRESS` | Verified node tags, `sys:gap-replay-batch-01` |

Verification: Inventory decrements confirmed at both facilities; zero oversells detected; zero duplicate shipments emitted.

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## 4.2 Leaky Bucket API Recovery Telemetry

Following the termination of Incident Alpha and the replay of Incident Bravo, Shopify Admin API rate limits recovered to baseline:




- **Recovery Rate:** Native drain rate of 2.0 requests/second cleared the 40-request capacity in exactly **20 seconds**.
- **Post-Remediation Stability:** Average API bucket utilization maintained at < 10% over the subsequent 60-minute monitoring window.

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## 5. Master Interconnectivity & System Architecture Lineage

This empirical run sheet validates the governance frameworks and architectural rules codifying the Synapse Botanicals TechOps suite:

| Upstream / Downstream Deliverable | Governing Node / Standard | Empirical Relationship & Proof-of-Execution |
|---|---|---|
| **SOP - OrderExceptionMutationProtocol - Parent (Domain 1.1)** | Hold Lifecycles & Mutation Gates | Verifies that automated tags (`cs:vip-tier-1`) do not mutate order fulfillment orders once they cross the Point of No Return. |
| **Matrix - OrderStatePointOfNoReturn - Master (Domain 1.2)** | State Transitions | Replay batch confirmed that all 14 backfilled orders were ingested in `OPEN / UNFULFILLED` state prior to 3PL allocation. |
| **Architecture - MultiNodeRoutingEngine - Spec (Domain 2.1)** | Regional Routing Node Allocation | Proves that backfilled orders containing Reno-exclusive SKUs (`SYN-TNCT-AM`, `SYN-ACC-WHSK`) were routed exclusively to `LOC-NV-02`. |
| **Matrix - OrderRoutingCostOptimization - DecisionEngine (Domain 2.2)** | Weight & Handling Costs | Validates that heavy baskets (`SYN-TMBL-MO`) continue to trigger packaging tags without loop interference. |
| **Matrix - CatalogDataSchemaAndValidation - Governance (Domain 3.2)** | Schema Integrity & UPCs | Confirms that verified SKUs and 12-digit UPC-A barcodes were mapped cleanly in the GraphQL backfill query. |
| **SOP - BusinessLogicAutomationAndWebhooks - Parent (Domain 4.1)** | Technical Standards & Kill-Switch | Empirically executes the 3-step Emergency Kill-Switch and verifies the 19-attempt retry failure model. |
| **Matrix - AutomationLogicAndWebhookTriage - Engine (Domain 4.2)** | Priority Ladder & Circuit Breaker | Proves the necessity of P0 priority gates and demonstrates how the `202 Accepted` queue pattern defeats the 5,000 ms webhook timeout. |